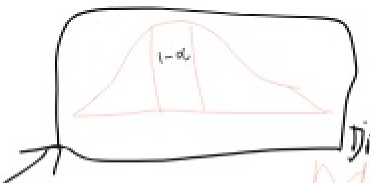
$H_0 : P_0 = 0{,}1$

$H_a = P_0 < 0{,}1$

$Z = 6{,}68$

Z kritis $= \boxed{-2{,}33}$



guna

Ditanya?

Ndek

$p = \frac{x}{n}$ → Jumlah sukses → Jumlah sample

$H_0 : P = P_0$

$H_a : P \neq P_0$

$P_0 = 10\% = 0{,}1$ $= \frac{28}{90}$

$$\boxed{Z = \frac{P - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}}} = \frac{\frac{28}{90} - \frac{1}{10}}{\sqrt{\frac{0{,}1(0{,}9)}{90}}}$$

$$= \frac{19/90}{\sqrt{\frac{0{,}09}{90}}}$$

$$= \frac{19/10}{\sqrt{\frac{9}{900}}}$$

$$= \frac{19/10}{10\sqrt{10}} =$$ 6,254

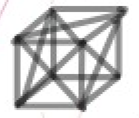
0,1
0,9
0,9
00
0,09

$\sqrt{1000}$
$= \sqrt{100 \cdot 10}$
$= 10\sqrt{10}$

$$\frac{\frac{19}{10}}{10\sqrt{10}}$$

$$\frac{19}{10} \times \frac{1}{10\sqrt{10}}$$

$$\frac{19}{100\sqrt{10}} =$$



$H_0 = P_0 = 0{,}1$ ②

$H_a = P_0 < 0{,}1$ → x

$Z_{hit} = \frac{P - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = 6{,}68$

$P_0 = 10\% = 0{,}1$ $P = \frac{28}{90} = 0{,}311$

$n = 90$

$x = 28$

$\alpha = 1\%$

$= 0{,}01$

$= -2{,}33$

1-α

terima H_0

bagus

Malah gambar

"Moana ... make WAY MAKE way"

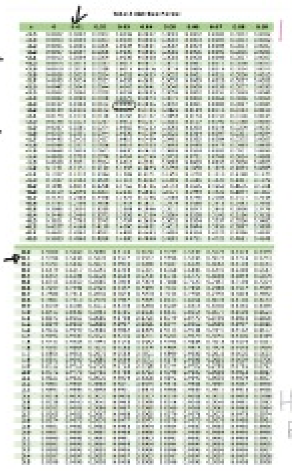
Interval kepercayaan

99% =

$100 - 99 = \boxed{1\%}$ → Taraf

Taraf

$-2{,}33$



P = Proporsi

< > ≠

Dit = ?

Rapi!

Sex

$$\boxed{p = \frac{x}{n}}$$

x → Jumlah ~~orang~~ sukses

n → Sample

$= \frac{28}{90} = 0{,}31$

$p = \frac{x}{n} = \frac{28}{90} = 0{,}31$ $P_0 = \frac{10}{100} = 0{,}1$

$\alpha = 1\% \to Z\ 0{,}01$

$$Z = \frac{P - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = \frac{0{,}31 - 0{,}1}{\sqrt{\frac{0{,}1(1-0{,}1)}{90}}} = \frac{0{,}21}{\sqrt{}}$$

$H_0 : P_0 = 28$

$H_1 : P\ 128$

$$S_P = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_P\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} =$$

$Z(0{,}01)$

Owl

Lia

dracrawl

2 sample hdep

$n_1 = 12$ $\bar{x}_1 = 80$ $\sigma = 4$

$n_2 = 10$ $\bar{x}_2 = 75$ $\sigma = 5$

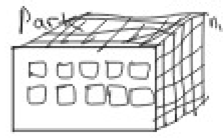
$n_1, n_2 < 20 \to t$

$n_1, n_2 \geq 20 \to Z$

$\boxed{H_0 : M_1 = M_2}$

$H_a : M_1 \neq M_2$

< > ≠



Binus?

sunib

~~$H_0 = M_1 = M_2$~~ $n_1, n_2 < 30 \to t$

$H_a\ M_1 \neq M_2$

$\bar{x}_1 = 80$ $\bar{x}_2 = 75$

$\sigma_1 = 4$ $\sigma_2 = 5$

$\alpha = 1\% = 0{,}01$ $db = 20$

$t = 0{,}01/2 = 0{,}005(db)$

$= 0{,}005(20) = 2{,}845$

terima H_0

$-2{,}845$

$$\frac{1}{12} + \frac{1}{10} = \frac{10+12}{120} = \frac{22}{120} = \frac{11}{60}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_P\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{5}{\sqrt{40{,}1}\sqrt{\frac{1}{12} + \frac{1}{10}}} = \frac{5}{\sqrt{40{,}1 \cdot \frac{11}{60}}} = \frac{5}{\sqrt{\frac{441}{60}}}$$

$$= \frac{5}{6{,}33\sqrt{\frac{1}{12} + \frac{1}{10}}}$$

$= \boxed{1{,}8447}$

$$S_P = \sqrt{\frac{(n_1-1)\sigma_1^2 + (n_2-1)\sigma_2^2}{n_1+n_2-2}}$$

$$= \sqrt{\frac{(11)(16) + (9)(25)}{12+10-2}} = \sqrt{\frac{176 + 225}{10}}$$

$$= \sqrt{\frac{401}{10}} = \sqrt{40{,}1}$$

$= 6{,}33$

25
9
225
176
401

5

$1 - 99\% = 0{,}01$

Terima

$\frac{0{,}01}{2}$ $\frac{0{,}01}{2}$

$-0{,}005$ $0{,}005$

$\bar{x}$ rata² $= 3{,}28$

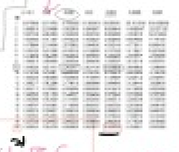
σ S $= 0{,}4185$

n = 10

$CI = \bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$

$S = \sigma$ 99 = 99,5

$dk = n - 1$
$= 10 - 1$
$= 9$



$t =$

$\bar{x} \pm t_{(\alpha/2)} \cdot \frac{s}{\sqrt{n}}$

$3{,}28 \pm 3{,}249 \cdot \frac{0{,}4185}{\sqrt{10}}$ 0,4308

batas atas = 3,28 + 0,4308 = 3,7108

batas bawah = 3,28 − 0,4308 = 2,85

$t\frac{\alpha}{2}, n-1$

b) $M_0 = 3{,}0$ $1 - 0{,}05$

$\alpha = 0{,}05$

$H_0\ M_0 = 3{,}0$

$H_a\ M_0 \neq 3{,}0$

Terima

$-\frac{0{,}05}{2}$ $\frac{0{,}05}{2}$

$-0{,}025$ $0{,}025 = 2{,}262$

≈ −2,262

2,12

−2,26 2,26

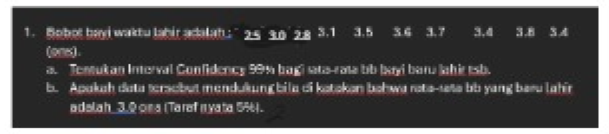
Langkah 4: Uji Hipotesis

interval kepercayaan $= 1 - \boxed{20}$

$= \boxed{20}$

0,5

$\alpha = 1 - 0{,}99 = 0{,}01/2 = 0{,}005(9)$



1. Bobot bayi waktu lahir adalah : 2.5 3.0 2.8 3.1 3.5 3.6 3.7 3.4 3.8 3.4 (ons).
   a. Tentukan Interval Confidency 99% bagi rata-rata bb bayi baru lahir tsb.
   b. Apakah data tersebut mendukung bila di katakan bahwa rata-rata bb yang baru lahir adalah 3.0 ons (Taraf nyata 5%).

$t = \boxed{0{,}01}/2$

$= \boxed{0{,}005}(9) =$ ~~3,250~~

3,250

a) $\bar{x} \pm \boxed{t\left(\frac{\alpha}{2}\right)} \frac{s}{\sqrt{n}} = 3{,}28 \pm 3{,}25 \frac{0{,}418}{\sqrt{10}}$

$3{,}28 \pm 0{,}429$

$3{,}85 \leq M \leq 3{,}701$

$-2{,}262$ $+2{,}262$

b)

(b) Uji Hipotesis

1. Hipotesis:
   - H_0: Rata-rata bobot bayi (μ) = 3.0
   - H_a: Rata-rata bobot bayi (μ) ≠ 3.0
2. Hitung statistik t:

$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$

$t = \frac{3.28 - 3.0}{\frac{0.418}{\sqrt{10}}} = 2.12$

3. Cari nilai p:
   Dari distribusi t ($df = 9$), nilai p untuk $t = 2.12$ adalah 0,063
4. Keputusan:
   Karena p = 0.063 > 0.05, gagal menolak H_0

9 0,05
0,025 =

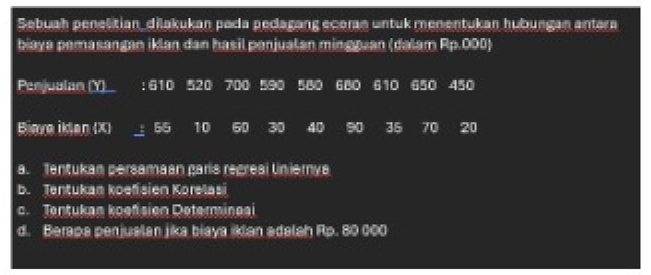
$\frac{0{,}05}{2} = 0{,}025$

$t = \frac{\bar{x} - M_0}{s/\sqrt{n}}$

$-0{,}025$ $+0{,}025$

$= \frac{3{,}28 - 3{,}0}{0{,}418/\sqrt{10}} = \boxed{2{,}12}$

4.



Sebuah penelitian dilakukan pada pedagang eceran untuk menentukan hubungan antara biaya pemasangan iklan dan hasil penjualan mingguan (dalam Rp.000)

Penjualan (Y)	: 610	520	700	590	580	680	610	650	450
Biaya iklan (X)	: 55	10	60	30	40	90	35	70	20

a. Tentukan persamaan garis regresi Liniernya
b. Tentukan koefisien Korelasi
c. Tentukan koefisien Determinasi
d. Berapa penjualan jika biaya iklan adalah Rp. 80 000

a) $y = a + bx = 403{,}62 + 2{,}53x$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = 2{,}53$$

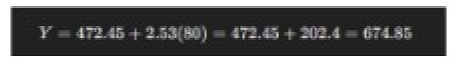
$a = \bar{y} - b(\bar{x}) = 598{,}89 - 2{,}53(45{,}56) = \boxed{403{,}62}$ X

b) $kk = r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}}} =$

c) $kp = r^2 \times 100\%$

d) $y = 403{,}62 + 2{,}53(80.000)$

$= 202.883{,}62$



$Y = 472.45 + 2.53(80) = 472.45 + 202.4 = 674.85$

Penjualan: Jika biaya iklan adalah Rp.80.000, maka penjualan diperkirakan Rp.674.850.